Assignment - 1 Solutions

(1) $$f(t) = DV_{in} + \sum_{n=1}^{\infty} \frac{2V_{in}}{n\pi} \sin(n\pi D) \cos(n\omega_s t - n\pi D)$$

where, $\omega_s = \frac{2\pi}{T_s}$

$\Rightarrow$ when, $D = \frac{1}{2}$, all the even harmonics vanishes.

(2) $$\frac{V_{out}(s)}{V_{sw}(s)} = \frac{\frac{1}{LC}}{s^2 + \frac{1}{R_L C}s + \frac{1}{LC}} = \frac{\omega_n^2}{s^2 + 2\xi\omega_n s + \omega_n^2}$$

$\Rightarrow$ natural frequency of oscillations $= \omega_n = \frac{1}{\sqrt{LC}} = 16,939.90892$ rad/sec.

$\Rightarrow$ damped frequency of oscillations $= \omega_d = \omega_n\sqrt{1-\xi^2} = 16939.34891$ rad/sec.

$\hookrightarrow$ for $\boxed{R_L = 5\Omega}$

$\Rightarrow$ frequency at which $\left|\frac{V_{out}(s)}{V_{sw}(s)}\right|$ peak occurs $= \omega\Big|_{Mpeak} = \omega_n\sqrt{1-2\xi^2} = 16938.78889$ rad/sec

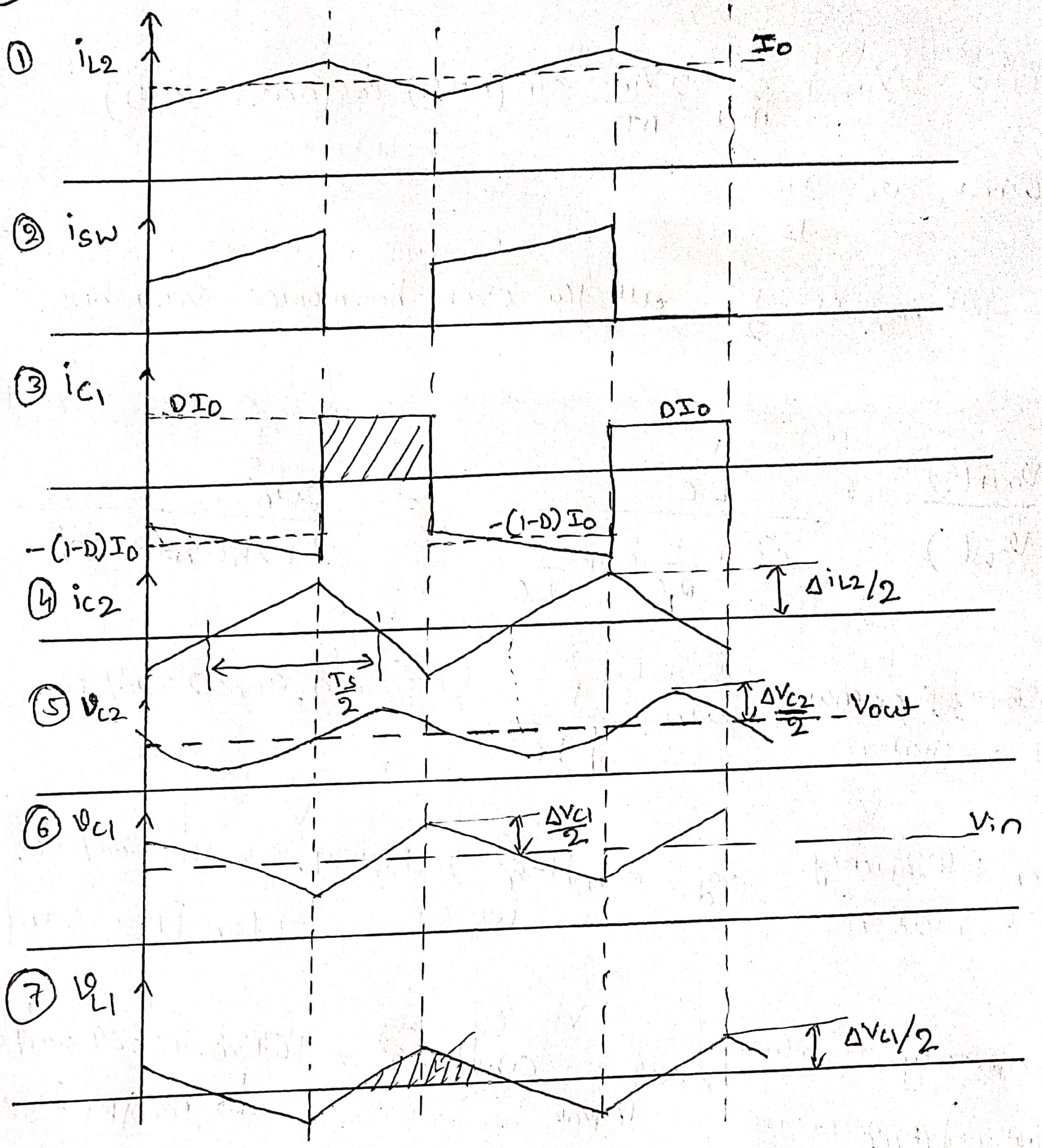
$\hookrightarrow$ for $\boxed{R_L = 5\Omega}$

$\Rightarrow$ Resonant frequency is a frequency at which the circuit behaves purely resistive $= \omega_{res} = \sqrt{\frac{1}{LC}\left[1 - \frac{L}{R^2C}\right]} = 16937.66878$ rad/sec

$\hookrightarrow$ for $\boxed{R_L = 5\Omega}$

(e) attenuation @ 200KHz is <u>-74.8135 dB</u>.

3



$$\boxed{L_2 = \frac{(V_{in} - V_{out})\,DT_s}{\Delta i_{L2}} = 75.5\,\mu H}$$

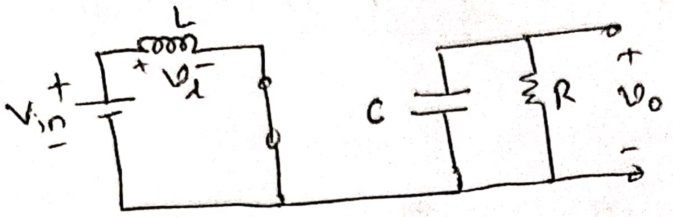
$$\boxed{C_1 = \frac{D(1-D)I_o T_s}{\Delta V_{C_1}} = 5.86\,\mu F}$$

$$\boxed{C_2 = \frac{\Delta i_{L2} \cdot T_s}{8\,\Delta V_{C2}} = 2.08334\,\mu F}$$

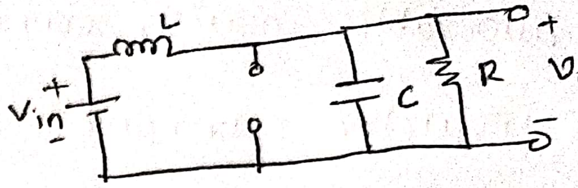
$$\boxed{L_1 = \frac{\Delta V_{C_1} T_s}{8\,\Delta i_{L_1}} = 60\,\mu H}$$

Scanned with OKEN Scanner

(5) DTs:-



(1-D) Ts



$$v_L = L\frac{di_L}{dt} = v_{in}$$

$$v_L = L\frac{di_L}{dt} = v_{in} - v_o$$

$$i_c = c\frac{dv_c}{dt} = \frac{-V_o}{R}$$

$$i_c = c\frac{dv_c}{dt} = i_L - i_o$$

$\Rightarrow$ applying Volt-sec balance equation, $\langle v_L \rangle = \frac{1}{T_s}\int_0^{T_s} v_L(t)\,dt = 0$

and also applying small ripple approximation.

$$\frac{1}{T_s}\left[V_{in} DT_s + (V_{in} - V_o)(1-D)T_s\right] = 0 \Rightarrow \boxed{V_o = \frac{V_{in}}{1-D}}$$

$\Rightarrow$ applying charge-balance equation, $\langle i_c \rangle = \frac{1}{T_s}\int_0^{T_s} i_c(t)\,dt = 0$

$$\frac{1}{T_s}\left[\left(\frac{-V_o}{R}\right) DT_s + \left(I_L - \frac{V_o}{R}\right)(1-D)T_s\right] = 0$$

$$\Rightarrow \boxed{I_L = I_{in} = \frac{I_o}{1-D}}$$

also, applying small ripple approximation.

Now, during DTs $\quad c\frac{dv_c}{dt} = \frac{-V_o}{R}$, we will look at the magnitude only.

$$\boxed{C = \frac{V_o DT_s}{R \Delta V_c}}$$

$\therefore$ at, $V_{in} = 5V \Rightarrow D = 0.6667 \Rightarrow \boxed{C = 88.8934\,\mu F}$

at, $V_{in} = 10V \Rightarrow D = 0.3334 \Rightarrow C = 44.4534\,\mu F$

Now, to calculate "L", it is mentioned that maximum allowable inductor current ripple is 20% of average inductor current.

average inductor current $= I_L = I_{in} = \frac{I_o}{1-D}$.

During DT_s $\Rightarrow$ $v_L = L\frac{di_L}{dt} = V_{in} \Rightarrow L = \frac{V_{in} D T_s}{\Delta i_L}$

W.K.T. $V_o = \frac{V_{in}}{1-D} \Rightarrow L = \frac{V_o D(1-D) T_s}{\Delta i_L}$

it is said that, $(\Delta i_L)_{max} = 20\%$ of $I_L = 20\%$ of $\left(\frac{I_o}{1-D}\right)$

$$\therefore L = \frac{V_o D(1-D) T_s}{0.2\left(\frac{I_o}{1-D}\right)} = D(1-D)^2 \frac{V_o T_s}{0.2 I_o}$$

$$\therefore \boxed{L = D(1-D)^2 \frac{V_o T_s}{0.2 I_o}}$$

at $\underline{V_{in} = 5V}$, $\Rightarrow I_{in} = 3A$, $\boxed{D = 0.6667}$, $\boxed{L = 111.1 \mu H}$

$\boxed{V_o = 15V}$ $\boxed{I_o = 1A}$ $\boxed{T_s = 20\mu s}$

at $V_{in} = 10V \Rightarrow I_{in} = 1.5A$, $D = 0.3334$, $\boxed{L = 222.22 \mu H}$

$V_o = 15V$, $I_o = 1A$, $T_s = 20\mu s$

so, if you use the value of $L = 222.22\mu H$ at all other input voltages b/w 5V & 10V the ripple % will be less that 20%.

let us, look at the equation, $L = D(1-D)^2 \frac{V_o T_s}{0.2\, I_o}$

let us generalize, this, instead of 0.2, i will take 'x'

where $0 \leq x \leq 1$

$$\therefore \quad L = D(1-D)^2 \frac{V_o T_s}{x I_o}$$

we can study, for a given value of 'L', at what 'D', x will be maximum, {$\because$ we are interested in maximum % ripple}

$$\therefore \quad x = D(1-D)^2 \frac{V_o T_s}{L I_o}$$

$$\frac{dx}{dD} = (3D^2 - 4D + 1) \frac{V_o T_s}{L I_o} = 0 \quad \{\text{To maximize}\}$$

$$\therefore \quad (D-1)(3D-1) = 0 \Rightarrow \boxed{D=1} \text{ or } \boxed{D=\frac{1}{3}}$$

you, can verify that $\boxed{D=\frac{1}{3}}$ is a maximum point

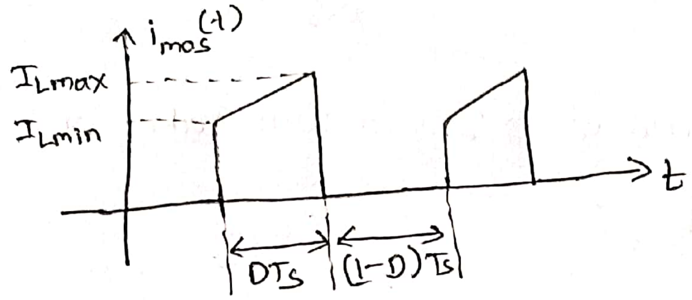
& $\boxed{D=1}$ is a minimum point.

$\therefore$ maximum ripple % will occur at $\boxed{D = 0.3334}$

which JUST happens to be in our case ~~lowest~~ highest input voltage, i.e. at $V_{in} = 10V$, $\boxed{D = 0.3334}$ for our case.

$\Rightarrow$ with changing input voltage the ~~dut~~ duty ratio and the average inductor current changes. The simple $D^*(1-D)$ will hold only when the average inductor current remains constant, and only the ripple will change due to duty ratio.

(d) Determination of rms value of Mosfet current.



$$i_{mos}(t) = \begin{cases} \dfrac{(I_{Lmax}-I_{Lmin})}{DT_s}\,t + I_{Lmin}; & DT_s \\ 0; & (1-D)T_s \end{cases}$$

$$I_{mos,rms}^2 = \frac{1}{T_s}\int_0^{DT_s}\left[\frac{(I_{Lmax}-I_{Lmin})}{DT_s}\,t + I_{Lmin}\right]^2 dt$$

$$I_{mos,rms}^2 = \frac{1}{T_s}\left[\frac{\left[\dfrac{(I_{Lmax}-I_{Lmin})}{DT_s}\,t + I_{Lmin}\right]^3}{3\,\dfrac{(I_{Lmax}-I_{Lmin})}{DT_s}}\right]_0^{DT_s}$$

$$I_{mos,rms}^2 = \frac{1}{T_s}\times\frac{DT_s}{3\,[I_{Lmax}-I_{Lmin}]}\left[\left[\frac{(I_{Lmax}-I_{Lmin})}{DT_s}\,.\,DT_s + I_{Lmin}\right]^3 - I_{Lmin}^3\right]$$

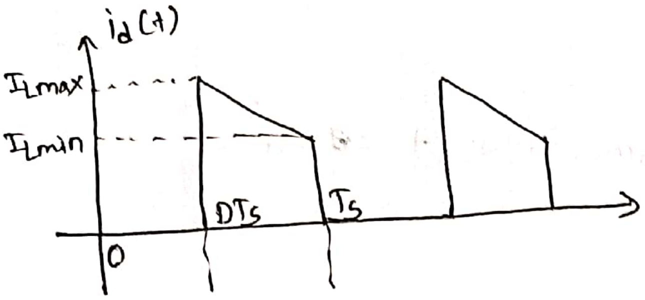
$$I_{mos,rms}^2 = \frac{D}{3(I_{Lmax}-I_{Lmin})}\left[\left[(I_{Lmax}-I_{Lmin}) + I_{Lmin}\right]^3 - I_{Lmin}^3\right]$$

$$I_{mos,rms}^2 = \frac{D}{3(I_{Lmax}-I_{Lmin})}\left[I_{Lmax}^3 - I_{Lmin}^3\right]$$

$$I_{mos,rms}^2 = \frac{D}{3(I_{Lmax}-I_{Lmin})}(I_{Lmax}-I_{Lmin})\left[I_{Lmax}^2 + I_{Lmin}^2 + I_{Lmax}I_{Lmin}\right]$$

$$\boxed{\therefore\ I_{mos,rms} = \sqrt{\frac{D}{3}\left[I_{Lmax}^2 + I_{Lmin}^2 + I_{Lmax}I_{Lmin}\right]}}$$

Determination of RMS value of Diode current



$$i_d(t) = \begin{cases} \dfrac{I_{Lmin} - I_{Lmax}}{(1-D)T_s}\, t + I_{Lmax} \ ; & DT_s < t < T_s \\ 0 \ ; & 0 < t < DT_s . \end{cases}$$

$$\Rightarrow I_{d,rms}^2 = \frac{1}{T_s}\int_{DT_s}^{T_s} \left[\frac{I_{Lmin} - I_{Lmax}}{(1-D)T_s}\, t + I_{Lmax}\right]^2 dt$$

let
$$\boxed{a = \frac{I_{Lmin} - I_{Lmax}}{1-D}}$$

$$\therefore I_{d,rms}^2 = \frac{1}{T_s}\int_{DT_s}^{T_s} \left[\frac{a}{T_s}\, t + I_{Lmax}\right]^2 dt$$

$$I_{d,rms}^2 = \frac{1}{T_s}\left[\frac{\left[\frac{a}{T_s}\, t + I_{Lmax}\right]^3}{3\left(\frac{a}{T_s}\right)}\right]_{DT_s}^{T_s}$$

$$I_{d,rms}^2 = \frac{1}{T_s} \times \frac{T_s}{3a}\left[\left[\frac{a}{T_s} \cdot T_s + I_{Lmax}\right]^3 - \left[\frac{a}{T_s} \cdot DT_s + I_{Lmax}\right]^3\right]$$

$$I_{d,rms}^2 = \frac{1}{3a}\left[(a + I_{Lmax})^3 - (aD + I_{Lmax})^3\right]$$

$$I_{d,rms}^2 = \frac{1}{3a}\Big[\left[a^3 + I_{Lmax}^3 + 3a^2 I_{Lmax} + 3a I_{Lmax}^2\right] - \left[a^3 D^3 + I_{Lmax}^3 + 3a^2 D^2 I_{Lmax} + 3aD I_{Lmax}^2\right]\Big]$$

$$I_{d,rms}^2 = \frac{1}{3a}\left[a^3(1-D^3) + 3a^2 I_{Lmax}(1-D^2) + 3a I_{Lmax}^2 (1-D)\right]$$

$$I_{d,rms}^2 = \frac{1}{3a}\left[\left[a^3(1-D)(1+D^2+D)\right] + \left[3a^2 I_{Lmax}(1-D)(1+D)\right] + 3aI_{Lmax}^2(1-D)\right]$$

$$I_{d,rms}^2 = \frac{(1-D)}{3}\left[a^2(1+D^2+D) + 3aI_{Lmax}(1+D) + 3I_{Lmax}^2\right]$$

<u>substitute for a</u>

$$I_{d,rms}^2 = \frac{(1-D)}{3}\left[\left[(I_{Lmin}-I_{Lmax})^2 \times \frac{\left[D^2+D+1\right]}{(1-D)^2}\right] + \left[3\frac{(I_{Lmin}-I_{Lmax})}{(1-D)} I_{Lmax}(1+D)\right] + 3I_{Lmax}^2\right]$$

$$\therefore \boxed{I_{d,rms} = \sqrt{\frac{(1-D)}{3}\left[\left[\frac{(I_{Lmin}-I_{Lmax})^2}{(1-D)^2}\left[D^2+D+1\right]\right] + \left[3\frac{(I_{Lmin}-I_{Lmax})I_{Lmax}(1+D)}{(1-D)}\right] + 3I_{Lmax}^2\right]}}$$